

# BRIGHTSIDE MOCK EXAMINATIONS

## COMPUTING

Essay and Objective

1 hour 45 mins

1&2



Name: \_\_\_\_\_

Index number: \_\_\_\_\_

Date: \_\_\_\_\_

## B.E.C.E 2026 BEST SPARK MOCK 2 – NOVEMBER, 2025

### COMPUTING [100 marks]

Write your **name** and **date** in **ink** in the spaces provided above

**DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.**

- ✓ This booklet consists of two papers, Paper 2 and paper 1
- ✓ Answer paper 2 in the booklet and paper 1 in the objective test answer sheet
- ✓ Paper 2 will last 1 hour.
- ✓ Paper 1 will last 45 mins.
- ✓ **Do not start paper 1 until you are told to do so**



@ 2025 Brightside Examination Consult

| For Examiners Use Only |       |
|------------------------|-------|
| Question Number        | Marks |
|                        |       |
|                        |       |
|                        |       |
|                        |       |
|                        |       |
|                        |       |
|                        |       |
| <b>TOTAL</b>           |       |

## PAPER 2 - ESSAY [60 MARKS]

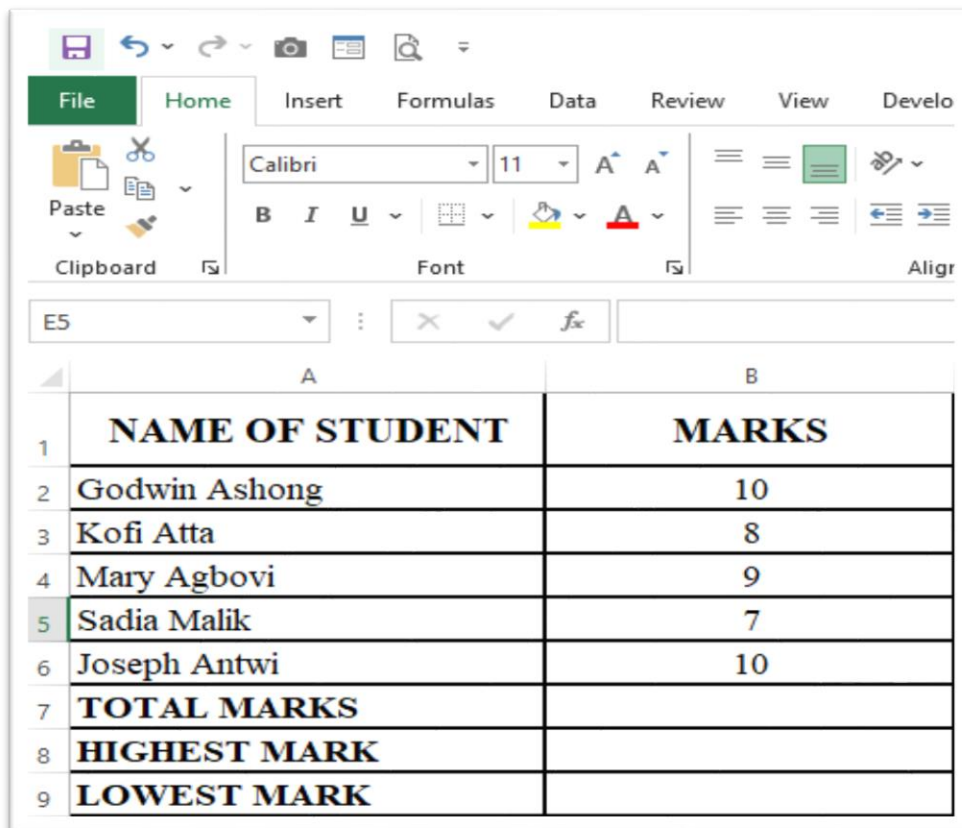
*This paper is in two sections A and B Answer Question 1 in Section A and any other three questions in Section B. Answer all the questions in the spaces provided in this question paper. At the end of the examination, you should submit the entire question paper to the invigilator*

### SECTION A

[24 MARKS]

Answer all questions in this section

1. The spreadsheet entries below represent marks obtained by 5 students in a class. Use the information to answer the questions below



The screenshot shows a Microsoft Excel spreadsheet with the following data:

|   | A               | B     |
|---|-----------------|-------|
| 1 | NAME OF STUDENT | MARKS |
| 2 | Godwin Ashong   | 10    |
| 3 | Kofi Atta       | 8     |
| 4 | Mary Agbovi     | 9     |
| 5 | Sadia Malik     | 7     |
| 6 | Joseph Antwi    | 10    |
| 7 | TOTAL MARKS     |       |
| 8 | HIGHEST MARK    |       |
| 9 | LOWEST MARK     |       |

(a) The name of the active cell is \_\_\_\_\_ [2marks]

(b) What is the name of the typeface used? \_\_\_\_\_ [2marks]

(c) Write a function in Microsoft Excel to determine the:

(i) highest mark; \_\_\_\_\_ [2marks]

(ii) lowest mark; \_\_\_\_\_ [2marks]

(iii) total mark; \_\_\_\_\_ [2marks]

(iv) average mark. [2marks]

(d) Describe how you would merge two or more table cells in Microsoft Word [2marks]

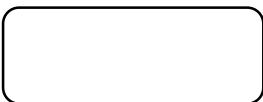
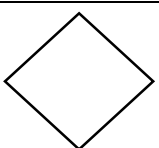
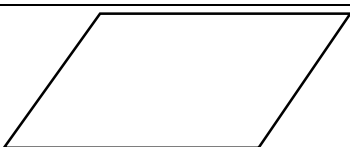
---

---

---

---

(d)State the function of the following flow diagrams in algorithm representation. [10marks]

| Diagram                                                                             | Function |
|-------------------------------------------------------------------------------------|----------|
|    |          |
|    |          |
|    |          |
|  |          |

**SECTION B**  
**[36MARKS]**

Answer *three* questions *only* from this section.

2(a)i. Parallel computing is on the rise in this information era, briefly explain parallel computing to your friends. [3marks]

---

---

---

---

ii. State three types of parallel computing. [3marks]

---

---

---

---

(b) The hard disk and solid state drive are both non-volatile storage, state any two differences between the hard disk and the solid state drive. [2marks]

---

---

---

---

---

---

---

---

(c) In the process of compressing file for an email communication, States the two main techniques that can be used for compressing files. [2marks]

---

---

---

(d) Assistive and adaptive technologies helps individuals with disabilities to ease their lives in society these days, as a computing student, differentiate between assistive technologies and adaptive technologies. [2marks]

---

---

---

---

---

---

---

---

3.(a)i. Your senior sister who is a business owner wants to advertise her products so that it goes far, suggest any three social media platforms she can use for her business advertisement. [3marks]

---

---

---

ii. Outline any two risks one is likely to encounter whilst he or she uses the social media. [1mark]

---

---

---

---

(b)i. Briefly explain what is meant by hard disk partitioning. [1mark]

---

---

---

ii. Outline any two importance of hard disk partitioning. [2marks]

---

---

---

(c)i. State any two features of the fifth generation computers that made them more powerful than the previous generation of computers. [1mark]

---

---

---

ii. Explain what is meant by artificial intelligence as used in the fifth generation of computers. [1mark]

---

---

---

---

(d)i. Word wrap is normally seen in application software such as word processes, explain what is meant by word wrap. [2marks]

---

---

---

ii. Differentiate between hard word wrap and soft wrap. [1mark]

---

---

---

---

4.a.i. Identify the spreadsheet data types in the table below;

| Date                         | Data type |
|------------------------------|-----------|
| = C3+D3-E3                   |           |
| =Product(B3:C3)              |           |
| 1 <sup>ST</sup> January 2025 |           |
| \$100.00                     |           |

[2marks]

ii. How will you differentiate between a spreadsheet function and a spreadsheet formula?  
[1mark]

---

---

---

(b)i. Folders and files are two things when it comes to the use of computers; state any two differences between a folder and a file. [1mark]

---

---

---

---

---

---

ii. Outline the steps involved in creating a folder without using a mouse. [1mark]

---

---

---

---

(c)i. Save and Save As commands are commonly used in most office applications, state the difference between Save and Save As command. [1marks]

---

---

---

---

---

ii. Provide the action that will be performed when the following keys are combined in the word processing application;

- a. Ctrl + C.....[ $\frac{1}{2}$  mark]
- b. Ctrl + K.....[ $\frac{1}{2}$  mark]
- c. Ctrl + V.....[ $\frac{1}{2}$  mark]
- d. Ctrl + D.....[ $\frac{1}{2}$  mark]

(d) In programming, operators and operands are fundamental concepts used to perform operations on data, differentiate between operators and operands. [1mark]

---

---

---

---

---

e.i. Your father once tuned in your television to the Ghana police TV and heard One police officer talk about data threat, since he's not all that inclined in information technology, he came to ask you what is meant by data threat, briefly explain what is meant by data threat to him. [1mark]

---

---

---

---

---

ii. State any two examples of data threats. [1mark]

iii. Outline any four ways by which data can be secured from threats. [2marks]

---

---

---

---

5(a)i. Outline the law of robotics as proposed by Isaac Asimov. [3marks]

---

---

---

---

---

---

---

---

---

ii. State any two disadvantages of the use of robots in everyday lives. [2marks]

---

---

---

bi. State the difference between transitions and animation as used in presentation programs. [1mark]

---

---

---

---

---

---

---

ii. State any two examples of transition effects in presentation programs. [1mark]

---

---

---

ci. Complete the table by writing the programming data types in the table below;

| Data                                                      | Data type |
|-----------------------------------------------------------|-----------|
| 1.28                                                      |           |
| “Ama”                                                     |           |
| A                                                         |           |
| -1                                                        |           |
| H.E John Dramani Mahama is the current president of Ghana |           |

[2marks]

ii. Evaluate  $5//2 + 4*2^3 + 6\text{mod}4$  [2marks]

---

---

---

---



PAPER 1 – OBJECTIVE TEST [40marks]

*Each question is followed by four options lettered **A** to **D**. Find out the correct option for each question and shade in pencil on your answer sheet the answer space which bears the same letter as the option you have chosen. Give only **one** answer to each question*

1. The basic unit of information in computing is:  
A. Byte  
B. Bit  
C. Kilobyte  
D. Megabyte
2. What is the main function of an operating system?  
A. To compile programs  
B. To manage computer hardware and software resources  
C. To facilitate internet browsing  
D. To perform arithmetic operations
3. Which of the following is an example of application software?  
A. Microsoft Word  
B. Linux  
C. BIOS  
D. UNIX
4. What device is primarily used to capture video input?  
A. Scanner  
B. Microphone  
C. Webcam  
D. Printer
5. Which of the following protocols is used for secure internet transactions?  
A. FTP  
B. HTTP  
C. HTTPS  
D. SMTP
6. Which device is used for entering text into a computer?  
A. Mouse

- B. Keyboard  
C. Monitor  
D. Printer
7. RAM in computing stands for:  
A. Read Access Memory  
B. Random Access Memory  
C. Read And Modify  
D. Random And Modify
8. Which of the following is a volatile memory type?  
A. ROM  
B. SSD  
C. RAM  
D. HDD
9. The part of the computer responsible for executing instructions is the:  
A. Hard drive  
B. CPU  
C. RAM  
D. Motherboard
10. The shortcut key to copy text in most software applications is:  
A. Ctrl + A  
B. Ctrl + S  
C. Ctrl + C  
D. Ctrl + X
11. Which of the following is not an operating system?  
A. Windows 10  
B. MacOS  
C. Linux  
D. Microsoft Word

12. What does WAN stand for in networking?

- A. Wide Area Network
- B. World Area Network
- C. Wireless Area Network
- D. Web Area Network

13. Which of the following is a primary function of an antivirus program?

- A. Increase processing speed
- B. Detect and remove malware
- C. Manage network connections
- D. Optimize software performance

14. In computing, what does URL stand for?

- A. Uniform Resource Locator
- B. Universal Resource Locator
- C. Uniform Retrieval Locator
- D. Universal Retrieval Locator

15. Which component is considered the "brain" of the computer?

- A. Hard drive
- B. CPU
- C. RAM
- D. Motherboard

16. Which of the following is a programming language?

- A. HTML
- B. HTTP
- C. USB
- D. ISP

17. The main function of a router is to:

- A. Store data
- B. Print documents
- C. Connect different networks
- D. Perform calculations

18. Which of the following is a disadvantage of cloud computing?

- A. Limited storage capacity
- B. Requires internet connection
- C. High hardware costs
- D. Limited accessibility

19. The software that allows users to browse the internet is called:

- A. Browser
- B. Search engine
- C. Web server
- D. ISP

20. A small, portable device used to store and transfer data is called:

- A. Hard drive
- B. SSD
- C. USB flash drive
- D. CD-ROM

21. Which of the following is not a characteristic of a good password?

- A. Short and simple
- B. Combination of letters, numbers, and symbols
- C. Unique and not easily guessable
- D. Regularly updated

22. The device used to connect a computer to a telephone line for internet access is called:

- A. Router
- B. Switch
- C. Modem
- D. Hub

23. The process of starting or restarting a computer is known as:

- A. Booting
- B. Logging in
- C. Installing
- D. Formatting

24. The command to create a new folder in most operating systems is:

- A. Ctrl + N
- B. Ctrl + F
- C. Ctrl + T
- D. Ctrl + O

25. Which of the following is not an input device?

- A. Scanner
- B. Mouse
- C. Printer
- D. Keyboard

26. The software used to create spreadsheets is:

- A. Microsoft Word
- B. Microsoft Excel
- C. Microsoft PowerPoint
- D. Microsoft Access

27. Which of the following is an output device?

- A. Monitor
- B. Keyboard
- C. Mouse
- D. Scanner

28. Which type of storage device is known for having no moving parts?

- A. HDD
- B. SSD
- C. Floppy disk
- D. DVD

29. In networking, what does LAN stand for?

- A. Local Access Network
- B. Local Area Network
- C. Large Area Network
- D. Long Area Network

30. Which of the following is a use of database management systems?

- A. Browsing the internet
- B. Managing large sets of structured data
- C. Editing images
- D. Designing websites

31. The process of converting human-readable data into a coded form is known as:

- A. Compression
- B. Encryption
- C. Decompression
- D. Decryption

32. Which device is used to convert printed documents into digital format?

- A. Printer
- B. Monitor
- C. Scanner
- D. Keyboard

33. The term used for a malicious software designed to harm, exploit, or otherwise compromise the integrity of a computer system is:

- A. Virus
- B. Program
- C. Software
- D. Application

34. Which of the following is an example of an optical storage device?

- A. USB flash drive
- B. Hard disk
- C. DVD
- D. SSD

35. Which component of a computer is responsible for generating the visual output on a display screen?

- A. Sound card
- B. Network card

- C. Graphics card
- D. Motherboard

36. Which of the following is an example of an input/output device?

- A. Printer
- B. Monitor
- C. Keyboard
- D. Touchscreen

37. What is the primary purpose of using a firewall in a computer network?

- A. To speed up the internet connection
- B. To protect the network from unauthorized access
- C. To store backup data
- D. To monitor system performance

38. Which type of memory is used to store the BIOS settings in a computer?

- A. RAM
- B. ROM
- C. Cache
- D. Flash memory

39. The process of finding and correcting errors in software code is known as:

- A. Compiling
- B. Debugging
- C. Executing
- D. Testing

40. Which of the following is a common use of spreadsheets in business?

- A. Writing reports
- B. Analyzing financial data
- C. Designing websites
- D. Sending emails



## SECTION A (OBJECTIVE TEST) [40marks]

|       |       |       |       |
|-------|-------|-------|-------|
| 1. B  | 11. D | 21. A | 31. B |
| 2. B  | 12. A | 22. C | 32. C |
| 3. A  | 13. B | 23. A | 33. A |
| 4. C  | 14. A | 24. A | 34. C |
| 5. C  | 15. B | 25. C | 35. C |
| 6. B  | 16. A | 26. B | 36. D |
| 7. B  | 17. C | 27. A | 37. B |
| 8. C  | 18. B | 28. B | 38. B |
| 9. B  | 19. A | 29. B | 39. B |
| 10. C | 20. C | 30. B | 40. B |

PAPER 2  
SECTION A

## QUESTION 1

1. a. E5 [2 marks]  
 b. Times New Roman [2 marks]  
 c. (i) highest mark;  
 =Max(B2:B6) [2marks]  
 (ii) Lowest mark;  
 =Min(B2:B6) [2marks]  
 (iii) Total mark;  
 =Sum(B2:B6) [2marks]  
 (iv) Average mark;  
 =Average(B2:B6) [2marks]

## c. Method 1: Using Table Tools

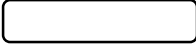
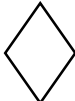
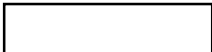
- Select the cells you want to merge by clicking and dragging across them.
- Go to the Table Layout tab (appears when the table is selected).
- Click Merge Cells in the Merge group.

## Method 2: Using Right-Click

- Right-click on the selected cells.
- Choose Merge Cells from the menu.

[Any one of the methods = 2 marks]

d.

| Diagram                                                                             | Function                                                                         |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|  | Used to represent Start/Stop stage of an algorithm                               |
|  | Used to represent decision making stage of an algorithm                          |
|  | Used to represent processing stage of an algorithm                               |
|  | Used to represent data input/output stage of an algorithm or programming process |

[2.5marks each]

## QUESTION 2

(a)i.

**Parallel computing** is a method where multiple processors or computers work together simultaneously to solve a problem faster by dividing tasks among them.

[3marks]

2(a)ii.

- Bit-level parallelism
- Instruction-level parallelism
- Data parallelism

[1mark each]

b(i)

- Hard disks use spinning magnetic platters, while SSDs use flash memory with no moving parts.
- SSDs are faster and more durable, while hard disks are generally cheaper and offer more storage capacity.

[1mark each]

c.

- Lossless compression
- Lossy compression

[1mark each]

d. **Assistive technologies** are tools designed specifically to help individuals with disabilities perform tasks they otherwise couldn't, while **Adaptive technologies** are modifications of existing tools to make them usable by individuals with disabilities.

[1mark]

## QUESTION 3

(a)i

- Facebook
- Instagram
- TikTok
- Twitter

[1mark for any 3]

(a)ii

- Cyberbullying
- Privacy breaches
- Identity theft

[0.5 marks each]

b(i) **Hard disk partitioning** is the process of dividing a hard drive into separate sections so each can function as an independent unit.

[1mark]

b(ii)

- It helps organize data and separate system files from personal files.
- It allows multiple operating systems to be installed on one disk.
- Data Protection
- Easy Backup and Recovery
- Improved Performance
- Simplified Disk Management
- Reducing Data Fragmentation

[1mark each for any 2]

c(i)

- Use of artificial intelligence
- High-speed processing and multitasking
- Natural Language Processing
- Superconducting Materials

[0.5marks each for any 2]

c(ii) **Artificial intelligence** refers to the ability of computers to perform tasks that typically require human intelligence, such as learning, reasoning, and problem-solving.

[1mark]

d(i) **Word wrap** is a feature in word processors that automatically moves text to the next line when it reaches the end of the current line.

[2marks]

d(ii) Hard word wrap inserts a line break manually, while soft wrap adjusts the text display without inserting a break.

[1mark]

#### QUESTION 4

a.i. Identify the spreadsheet data types in the table below;

| Date                         | Data type |
|------------------------------|-----------|
| = C3+D3-E3                   | Formula   |
| =Product(B3:C3)              | function  |
| 1 <sup>ST</sup> January 2025 | Date      |
| \$100.00                     | Currency  |

[0.5marks each]

ii. How will you differentiate between a spreadsheet function and a spreadsheet formula?

A **spreadsheet function** is a predefined formula that performs a specific calculation or operation while a **spreadsheet formula** is a user-defined expression that calculates a value using values, cell references, and operators. Functions are typically part of formulas.

[1mark]

bi.

- A folder can contain another folder whereas a file cannot contain another file.
- Folders do not have name extension whereas files have name extension.
- Folders do not need application software to open whereas files need application software to open.

[0.5marks each for any 2]

ii.

- Navigate to the desired location using keyboard (e.g., arrows, Tab).
- Press Ctrl + Shift + N to create a new folder.
- Type the folder name and press Enter.

[1mark]

c.i. Save and Save As commands are commonly used in most office applications, state the difference between Save and Save As command.

**Save** updates the current file with changes while **Save As** lets you save the file with a new name, location, or format.

[1mark]

ii. Provide the action that will be performed when the following keys are combined in the word processing application:

- a. Ctrl + C – Copy selected text.
- b. Ctrl + K – Insert hyperlink
- c. Ctrl + V – Paste copied text/ item.
- d. Ctrl + D – Open font dialog box

[0.5 marks each]

d. Differentiate between operators and operands in programming.

**Operators** are symbols or keywords used to perform operations on data (like +, -, \*, /) while **Operands** are the values or variables that these operators act upon.

[1mark]

e.i. Explain what is meant by data threat.

A **data threat** refers to a potential occurrence that could compromise the security, integrity, or availability of data. This could lead to unauthorized access, theft, loss, or damage to the data.

[1mark]

e.ii. State any two examples of data threats.

- Hacking: Unauthorized access to data by malicious actors.
- Insider threat
- Natural disaster.
- Malware: Software designed to harm or exploit data.

[0.5marks each]

e.iii. Outline any four ways by which data can be secured from threats.

- **Encryption:** Protecting data by converting it into a coded form.
- **Access Controls:** Limiting who can access or modify data.
- **Backups:** Creating copies of data to prevent loss.
- **Firewalls:** Blocking unauthorized access to systems holding data.

[0.5marks each]



## QUESTION 5

a.i. Outline the law of robotics as proposed by Isaac Asimov.

**Isaac Asimov's Three Laws of Robotics states that;**

- A robot may not injure a human being or, through inaction, allow a human being to come to harm.
- A robot must obey the orders given it by human beings except where such orders would conflict with the First Law.
- A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

[3marks]

5a.ii. State any two disadvantages of the use of robots in everyday lives.

- Job Displacement
- Over-reliance on robots can lead to loss of skills and decreased human interaction.
- Purchasing and maintaining robots can be expensive.
- Robots can malfunction or be programmed incorrectly, posing risks to humans.
- Robots may struggle with tasks that require flexibility or complex decision-making.
- Increased use of robots could lead to reduced human interaction and social isolation.
- Robots can be vulnerable to hacking and cyber-attacks.

[1mark each for any 2]

bi. Difference between transitions and animation in presentation programs:

**Transitions** are effects that happen between slides when moving from one slide to another while **Animations** are effects applied to elements (like text, images) on a slide to make them move or change.

[1mark]

ii. Two examples of transition effects in presentation programs:

- Fade in/out
- Slide push
- Dissolve
- Wipe.
- Zoom
- Flip:
- Rotate
- Morph

[0.5marks each for any 2]

ci. Programming data types in the table:

| Data                                                      | Data type |
|-----------------------------------------------------------|-----------|
| 1.28                                                      | Float     |
| “Ama”                                                     | String    |
| A                                                         | Character |
| -1                                                        | Integer   |
| H.E John Dramani Mahama is the current president of Ghana | Boolean   |

[2marks]

ii. Evaluate  $5//2 + 4*2^3 + 6\text{mod}4$ :

Following order of operations:

1.  $2^3 = 8$
2.  $4*8 = 32$
3.  $5//2 = 2$
4.  $6\text{mod}4 = 2$
5.  $2 + 32 + 2 = 36$

[2marks]